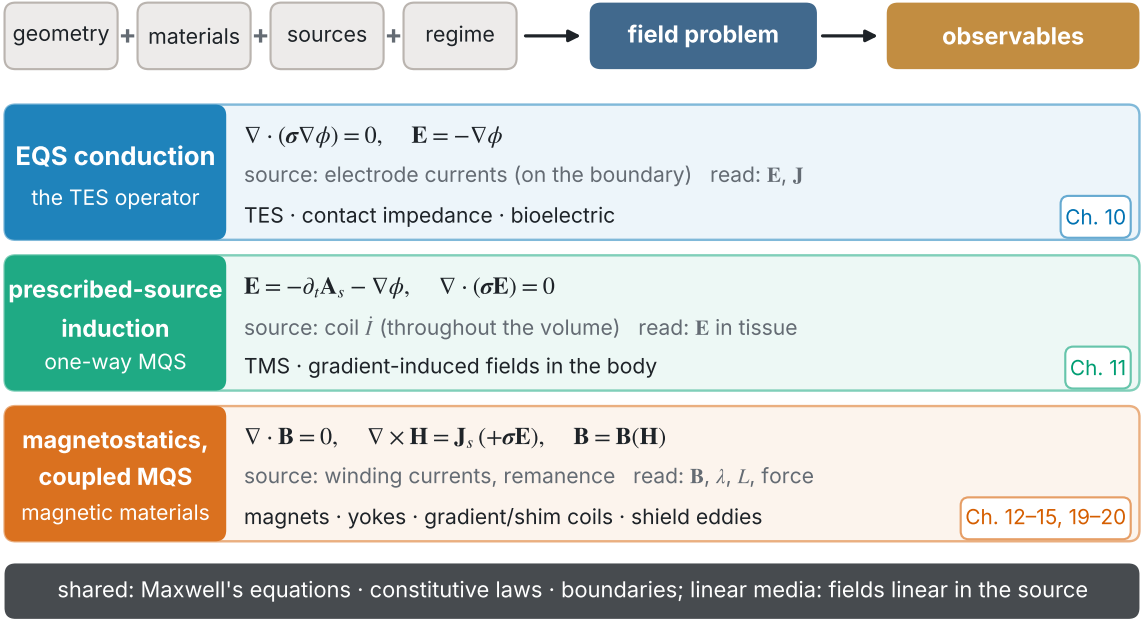
geometry + materials + sources + regime
field problem
observables
EQS conduction
the TES operator
$\nabla \cdot (\boldsymbol{\sigma} \nabla \phi) = 0, \quad \mathbf{E} = -\nabla \phi$
source: electrode currents (on the boundary) read: $\mathbf{E}$, $\mathbf{J}$
TES · contact impedance · bioelectric
Ch. 10
prescribed-source induction
one-way MQS
$\mathbf{E} = -\partial_t \mathbf{A}_s - \nabla \phi, \quad \nabla \cdot (\boldsymbol{\sigma} \mathbf{E}) = 0$
source: coil $\dot{I}$ (throughout the volume) read: $\mathbf{E}$ in tissue
TMS · gradient-induced fields in the body
Ch. 11
magnetostatics, coupled MQS
magnetic materials
$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{H} = \mathbf{J}_s \, (+\boldsymbol{\sigma} \mathbf{E}), \quad \mathbf{B} = \mathbf{B}(\mathbf{H})$
source: winding currents, remanence read: $\mathbf{B}$, λ, L, force
magnets · yokes · gradient/shim coils · shield eddies
Ch. 12–15, 19–20
shared: Maxwell's equations · constitutive laws · boundaries; linear media: fields linear in the source